

T. Pushpalatha · P. Ramachandra Reddy ·
P. Sreenivasula Reddy

Impairment of male reproduction in adult rats exposed to hydroxyprogesterone caproate in utero

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Abstract Hydroxyprogesterone caproate is one of the most effective and widely used drugs for the treatment of uterine bleeding and threatened miscarriage in women. Hydroxyprogesterone caproate was administered to pregnant rats in order to assess the effect of intraperitoneal exposure to supranormal levels of hydroxyprogesterone caproate on the male reproductive potential in the first generation. The cauda epididymal sperm count and motility decreased significantly in rats exposed to hydroxyprogesterone caproate during embryonic development, when compared with control rats. The levels of serum testosterone decreased with an increase in follicle stimulating hormone and luteinizing hormone in adult rats exposed to hydroxyprogesterone caproate during the embryonic stage. It was suggested that the impairment of male reproductive performance could be mediated through the inhibition of testosterone production.

Introduction

Recently, considerable concern has been expressed over the possibility that exposure to excess female hormone or other oestrogenic chemicals may adversely affect male reproduction in wildlife and humans. There is evidence of a significant fall in average sperm count (Carlsen et al. 1992) together with a progressive increase in incidence of testicular cancer (Kavlock et al. 1996) in normal men over the past 50 years. The available data have also suggested an increased incidence of certain human male reproductive tract abnormalities such as cryptorchidism and hypospadias during the same period (Toppari et al. 1996). It

has been hypothesized that all the above male reproductive abnormalities are interrelated and that they may have a common origin in embryonic and/or neonatal life. One ironic example is that the children of women treated with diethylstilbestrol (a synthetic oestrogen that was formerly used by physicians as an anti-abortion agent) are now suffering from reproductive abnormalities (Takasugi and Bern 1988; Hines 1992).

In view of this, an elaborate programme to evaluate the effect of transplacental exposure to supranormal levels of female hormones on reproductive health in male rats was initiated. We have reported earlier that exposure to hydroxyprogesterone caproate in utero caused a significant decrease in steroidogenic enzyme activity levels in the testis of male rats (Pushpalatha et al. 2003). The present report is the second in that series and examines the levels of testosterone, follicle stimulating hormone (FSH) and luteinizing hormone (LH) in the serum of adult rats exposed to supranormal levels of hydroxyprogesterone caproate during embryonic development. The sperm count and percentage motility from cauda epididymis were also studied in both control and experimental rats.

Materials and methods

Wistar rats bred in our own animal facility were maintained under a controlled light:dark (12:12 h) schedule at 23±1°C and provided food and water ad libitum. The rat feed was purchased from Kamadhenu Agencies, Bangalore, India. Only adult rats (90 days old) were used in the present study.

Hydroxyprogesterone caproate (trade name: Proluton depot) manufactured by German Remedies, Goa, was purchased from a local pharmacy. This drug is most commonly prescribed to control uterine bleeding and to protect threatened pregnancies in women. Hydroxyprogesterone caproate (250 mg in 1.0 ml) is available in an oily solution of castor oil IP and benzyl benzoate IP (1:1.7).

Rats were allowed to mate and pregnant rats were isolated after identification using the vaginal smear technique. Pregnant rats were divided into three groups of six animals each. The first group, which served as the control, was treated the same as the experimental group but received injections of a mixture of castor oil and benzyl benzoate (1:1.7) in 15 µl volume. Groups 2 and 3 received injections (10 mg/kg body weight or 25 mg/kg body weight per injection, respectively) of hydroxyprogesterone caproate. Injections

T. Pushpalatha · P. Ramachandra Reddy ·
P. Sreenivasula Reddy (✉)
Department of Biotechnology,
Sri Venkateswara University,
517 502 Tirupati, India
e-mail: reddy_1955@yahoo.co.in
Tel.: +91-877-2249320
Fax: +91-877-2249611

Table 1 Changes in epididymal sperm count and motility in adult rats exposed to hydroxyprogesterone during embryonic development

Parameter	Control	Hydroxyprogesterone injected	Hydroxyprogesterone injected
		(10 mg/kg body weight)	(25 mg/kg body weight)
Sperm count (million/ml)	50±6.3	38*±2.3 (-24.00%)	27*±1.6 (-46.00%)
Motility (%)	77±2.6	48*±0.9 (-37.66%)	34*±1.7 (-55.84%)

Values are mean ±SD of six individual rats

Values are significantly different from controls (* P <0.001)

Table 2 Changes in testosterone, FSH and LH levels in serum of adult male rats exposed to hydroxyprogesterone during embryonic development

Hormone	Control	Hydroxyprogesterone injected	Hydroxyprogesterone injected
		(10 mg/kg body weight)	(25 mg/kg body weight)
FSH (ng/ml)	6.25±0.28	10.31*±0.16 (64.96%)	18.88*±0.55 (202.08%)
LH (ng/ml)	1.40±0.01	1.93*±0.05 (37.85%)	7.54*±0.28 (438.57%)
Testosterone (ng/ml)	9.90±0.79	6.88**±0.88 (-30.50%)	5.25*±0.52 (-46.97%)

Values are mean ±SD of six individual animals

Values are significantly different at * P <0.001; ** P <0.001

were administered intraperitoneally to pregnant rats on the 1st, 7th and 14th day of pregnancy.

The male pups from all three groups were allowed to grow on a normal diet. After 90 days, the male rats (180±10 g body weight) were decapitated and the blood was collected by cardiac puncture. The serum was separated by centrifugation at 2,000 g for 15 min after overnight storage at 4°C and then stored at -20°C for hormone assays. Epididymis was excised at autopsy and placed in 10-ml vial containing 1.0 ml of physiological saline at 37°C. A suspension of spermatozoa was prepared by mincing with a razor blade on a glass slide in a few drops of physiological saline. A drop of the resulting sperm suspension was transferred to a microscopic slide for analysis of total sperm count and motility (Belsey et al. 1980).

A radioimmunoassay of serum testosterone was performed by the method of Rao et al. (1990). Serum FSH and LH levels were determined according to the method of Lin et al. (1988). Iodination of rFSH and rLH with ^{125}I was performed by the method of Greenwood et al. (1963) using chloramine-T as an oxidizing agent. The data were analysed statistically using Student's t -test (Pillai and Sinha 1968).

Results

The effect of hydroxyprogesterone caproate on cauda epididymal sperm count and motility in rats exposed to the hormone in utero are given in Table 1. A significant decrease in sperm count ($t=23.46$; $df=10$; $P<0.001$) and reduction in motility ($t=17.76$; $df=10$; $P<0.001$) was observed in rats exposed to hydroxyprogesterone caproate prenatally when compared with control rats. The changes were greater in rats exposed to higher doses of hydroxyprogesterone caproate. The levels of serum testosterone were significantly decreased ($t=6.21$; $df=10$; $P<0.001$) in adult rats exposed to hydroxyprogesterone caproate during embryonic development when compared with the corresponding group of control rats (Table 2). However, the levels of serum FSH and LH increased significantly ($t=27.56$; $df=10$; $P<0.001$ and $t=18.89$; $df=10$; $P<0.001$ respectively) in rats exposed to hydroxyprogesterone caproate in utero when compared with the corresponding group of control rats (Table 2).

Discussion

Diethylstilbestrol (DES) is a synthetic oestrogen which was used by physicians to prevent spontaneous abortions in women for more than 25 years until its use for this purpose was banned in the early 1970s because of it was believed to cause transgenerational reproductive abnormalities. The transplacental effects of DES, a synthetic therapeutic agent, on the fetuses of both humans and experimental animals have been confirmed. Daughters born to mothers who took diethylstilbestrol now suffer increased rate of vaginal adenocarcinoma, various genital tract abnormalities and abnormal pregnancies. Children exposed in utero to DES experience congenital abnormalities of their reproductive system and reduced fertility (Gill 1976; McLachlan et al. 1980, 1982; Newbold et al. 1991; Bern 1992). In humans, the DES model is clear and traceable. Hydroxyprogesterone caproate is one of the most widely used anti-abortive drugs in this part of the country.

The results show that exposure to hydroxyprogesterone caproate during embryonic development significantly affects male reproduction. Prenatal exposure to hydroxyprogesterone caproate significantly decreased the epididymal sperm count and sperm motility in rats. The data also reveal elevated FSH and LH levels, associated with significantly reduced circulating testosterone levels in rats exposed to hydroxyprogesterone caproate during embryonic development, indicating a probable inhibition of androgen synthesis. This decrease in testosterone titres could be due to the diminished responsiveness of Leydig cells to LH and/or the direct inhibition of testicular steroidogenesis. We observed a significant decrease in the activity levels of 3β -hydroxysteroid dehydrogenase and 17β -hydroxysteroid dehydrogenase in the testes of rats exposed to hydroxyprogesterone caproate in utero (Pushpalatha et al. 2003). Thus the decreased serum testosterone levels may be due to decreased steroidogenesis in rats exposed to hydroxyprogesterone caproate during embryonic development. Bern (1992) reported reproduc-

tive abnormalities in adult rats exposed to oestrogen during the embryonic stage and in the first few days after birth and concluded that the changes, once induced, were oestrogen-independent. He also concluded that the long-term permanent changes in adult rats as a result of exposure to oestrogen during development can occur without apparent birth defects in the neonate. This deficiency in androgen levels of exposed rats led to an androgen deprivation effect on target organs. It is well known that androgens are essential for normal spermatogenesis (Sharpe 1987).

The increased levels of LH, together with decreased serum testosterone levels, indicate decreased steroidogenic activity of the testis in the experimental rats. The increase in serum FSH levels indicates impairment of spermatogenesis in the experimental rats and reflects the germ-cell loss or damage to the Sertoli cells, thereby affecting the feedback regulation of FSH secretion (Van Thiel et al. 1972). The elevated levels of serum FSH and LH with lowered testosterone levels in the experimental rats are indicative of an intact pituitary–testicular axis.

These results provide the first direct evidence that exposure to hydroxyprogesterone caproate during embryonic development impairs the titre of serum FSH, LH and testosterone in adult rats. The decreased testosterone levels might also be responsible for the decreased sperm count and sperm motility in the experimental rats. It is also suggested that the observed impairment of male reproductive performance could be mediated through the inhibition of testosterone production in hydroxyprogesterone caproate-exposed rats, which is essential to maintain spermatogenesis quantitatively. The data reported in the present study demonstrate that prenatal exposure to relatively low doses of hydroxyprogesterone caproate can adversely affect the fertility of male offspring. Extrapolation of rat data to human being is always difficult. However, it should be noted that the highest dose used in this study (25 mg/kg body weight) did not differ markedly from that often administered therapeutically to pregnant women.

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